



Week 1 Lecture 5

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	5
▼ Type	Lecture
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Introduction to the Web

- Platform of choice for this course
- Generic - works across operating systems, hardware architectures
 - Can it be considered as a cross-platform OS?
- Built on sound underlying principles
- Worth understanding
 - Constraints: what can and cannot be done (easily)
 - Costs: storage, network, device sizing, datacenter

Historical background

- Telephone networks ~ 1890+
 - Circuit switched - allow A to talk to B, through complex switching network
 - Physical wires tied up for duration of call even if nothing was said!
 - The bandwidth used by voice signal is pretty low
- Packet switched networks ~ 1960s
 - Wires occupied only when data was to be sent - more efficient use

Last link: The physical connection between my telephone and my nearest telephone exchange, it has to be tied up to me

Aggregate all the data in one place (telephone exchange) and transmit it over a single wire at a much higher speed (connection between telephone exchanges)

- Data is being sent instead of voice over the wires
- ARPANet - 1969
 - acronym for Advanced Research Projects Agency Network
 - Node-to-node network
 - Mostly university driven
- Others:
 - IBM SNA, Digital DECNet, Xerox Ethernet, ...
- Protocol
 - How to format packets; place them on a wire; headers / checksums;
 - Each network had its own protocol
- "Inter" network
 - How to communicate between different network protocols?
 - Or replace them with a single "inter"net protocol
- IP: Internet Protocol - 1983
 - Define headers, packet types, interpretation
 - Can be carried over different underlying networks: ethernet, DECnet, PPP, SLIP
- TCP: Transmission Control Protocol - 1983
 - Establish reliable communication - retry; error control
 - Automatically scale and adjust to network limits
 - It is a connection oriented protocol
- Domain names ~ 1985
 - Use names instead of IP addresses
 - IP addresses were basically a set of 4 numbers, what they called the dotted quad notation
 - There are 4 numbers in the range of 0 - 255, which are separated by dots (.)
 - Easy to remember - *.com* revolution still in the future
- HyperText ~ 1989+
 - Text documents to be "served"
 - Formatting hints inside documents to "link" to other documents - HyperText
 - It was brought into one consolidated framework by Tim Berners Lee at CERN
 - All this lead to the creation of the World Wide Web

Where are we now?

- Original web was limited to ...
 - static pages
 - complicated executable interfaces
 - limited styling
 - browser compatibility issues
- Web 2.0 ~ 2004+
 - dynamic pages - generated on the fly
 - HTTP as a transport mechanism - binary data; serialized objects
 - client side computation and rendering
 - platform agnostic operating system